EMERGING TECHNOLOGIES OVERVIEW AND ITS AREAS

Comprehensive Report

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INTRODUCTION

Technology has become a big part of our lives, whether inside our house or in the outside world. Technology continuously influences us people on the way we live and work. Every single day there are new developments that can improve existing technology and introduce new ways of performing tasks. Some may create new opportunities for every industry like in business and other organizations. Technologies have become established, while others are continuing to develop and considering emerging technologies.

In this module, we are talking about Emerging Technologies and Its Areas, wherein it focuses on understanding how technologies nowadays emerge in the real world and how they can be potentially used in different industries. This module aims to help every learner to understand and explain emerging technology, describe its areas, and analyze several emerging technologies.

The concept of this topic is wider than simply understanding a new invention. Some technologies may have existed in the past but still can be considered as emerging technologies because they are continuously evolving and being used in a new application.

Understanding emerging technologies is an important matter because it allows individuals and organizations to recognize and consider its possible effect on industries and society.

I. OVERVIEW OF EMERGING TECHNOLOGIES

According to BusinessDictionary.com emerging technologies is a technology that is currently developing and expected to be developed within the next five to ten years and it may change the business and social environment.

This module defines that it does not completely describe emerging technology. Not every new emerging technology is always accurate. Every technology can be also called emerging through advancement of an existing innovation or through simple application to modern problems. For example, Artificial Intelligence (AI) was initially developed in the 1950s. Regardless of its long history, it still remains as an emerging technology because it continues to progress and it is also being introduced into different fields and industries.This is how emerging technologies demonstrates continual progress. It has become an important part of our everyday lives because of its improvements and capabilities.

According to Rotolo, Hicks, and Martin (2015) said that emerging technologies can be understood through five major characteristics:

I. Radical novelty

This refers to the innovation or substantial difference that is associated with an emerging technology. Novelty does not always mean that technology was not recently invented. An existing technology can become novel when we are using it for a new purpose or can be combined with other technologies in some ways.

It also can be radically novel when it is performing a particular function using a different basic principle from an old technology that has the same functionality.

For example, AI proves this characteristic because the application of AI is continuously expanding. Though AI has existed before, its use for different purposes creates new possibilities and outcomes. Hence, radical novelty can come from either the development of a new technology approach or original application of an existing technology.

II. Relatively fast growth

Emerging technologies generally experience growth compared to the technologies that are already available. Growth can be measured in terms of:

- The number of users
- The number of companies, institutions, and researchers involved
- Public and private funding
- Prototypes
- Services
- Products

Artificial Intelligence is an example of relatively fast growth, because AI is increasingly used by many companies, institutions and individuals. This growth demonstrates that an emerging technology can acquire momentum as more people and companies begin researching, developing and using it.

III. Coherence

This occurs when earlier separate research areas and technologies become connected. For an emerging technology to display coherence, the technologies involved must be moved over purely theoretical ideas and develop connections practically. Over the time these connections remain, the technology earns a stronger direction and identity.

IV. Prominent Impact

This pertains to the important impact that an emerging technology can have on industries and society. The impact may be restricted to a specific area or may extend over several sectors. An emerging technology can influence institutions and organizations, communication between people and systems, and the way knowledge is created.

Example is Artificial Intelligence because according to the module, AI can create significant changes in:

- Business
- Institutions
- Healthcare
- Individual

This shows that the impact of an emerging technology can extend over a single industry.

V. Uncertainty and Ambiguity

Uncertainty exists because of the most significant effects of an emerging technology that may occur in the future. It is hard to know when will the application will succeed or what the final effects of technology will be.

Ambiguity exists because it is possible for the application of an emerging technology to still be changed. They can also have different interpretations of what technology means or how it can be used.

Another example for Artificial Intelligence, people can make many predictions in AI applications but these predictions do not have assurance that the application will actually be developed successfully. Therefore, emerging technologies should be approached with caution and interest because their future development is not 100% predictable.

II. AREAS OF EMERGING TECHNOLOGY

1. Augmented Reality (AR)

This combines the real world with visual effects, text, and sound that are called computer-generated imagery. Examples for this given in the module are the in-car parking guides, Pokemon GO, and applications that allows shoppers to visualize how the furniture would look in their homes

Somehow, AR can also be used in work to provide workers with real-time information, such as selection for warehouse parts and repair instructions.

2. Artificial Intelligence

This is the imitation of human intelligence by machines. This includes process of learning, reasoning, and self-regulation.

Even though AI was developed as early as the 1950s, it still remains as an emerging technology because it is continuously developing and applied in many areas.

The impact of AI can be seen in many fields such as businesses, universities, healthcare, and among individuals.

3. Machine Learning

This is an application of AI wherein allows the machine to learn from experience . The module identifies healthcare, financial services, marketing and transportation as areas where machine learning can be applied.

In healthcare, machine learning can help identify and develop new medicines swiftly.

4. Quantum Computers

This uses quantum mechanics to perform various calculations more effectively than conventional computers. The basic unit is called qubit, it is different from a traditional bit because it can exist in a quantum superposition of two computational states.

This module mentions IBM, Google, Rigetti Computing, and IonQ. The future of quantum computers may be able to stimulate atoms and molecules and assist in designing materials.

III. THE FOUR INDUSTRIAL REVOLUTIONS

STAGES	MAIN DEVELOPMENT
INDUSTRY 1.0	Mechanical production using hydroelectric and steam-powered equipment

INDUSTRY 2.0	Electricity enabled greater production and more polished machines
INDUSTRY 3.0	Computers, electronics and other technology raised automation.
INDUSTRY 4.0	Linked technologies alter work and productions

Currently Industry 4.0 is still a developing industrial environment. It changes relationships between producers, customers and humans and machines.

IV. Building Blocks of Industry 4.0

Big Data Analytics

This involves collecting an amount of data looking at it carefully and putting it in order so that trends and useful information can be found. I see that Big Data Analytics helps people make decisions about customer service and production.

Internet of Things

This connects objects to the Internet so that they can talk to each other. I notice that the Internet of Things lets sensors and wireless technologies share data and give answers

Cloud Computing

This delivers services that are hosted on the Internet. Observing that Cloud Computing enables production services that rely on data because more machine data and functions are moved to the cloud as Industry 4.0 grows.

Augmented Reality

This gives employees real-time information on devices such as phones. I find that Augmented Reality helps with warehouse work and supplies repair instructions improving work processes and decision-making.

V. Conclusion

All newly developed technologies are not only a part of emerging technology. Once a technology starts evolving, it can have a big impact on the future even its uncertain, but it can still be considered emerging. Overall, new technologies have the potential to help and change industries and society. Understanding their benefits is important when considering their future applications.

VI. REFERENCES

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AI Detection results

2% of this text appears to be AI-generated



Resembles AI text	2%
No AI text patterns found	98%

Your Text is Human written

